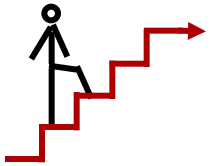
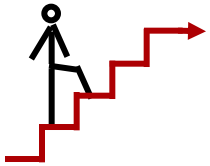


# Working with Databases using JDBC



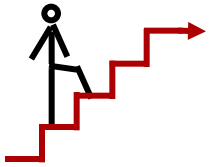
# JDBC Drivers

- JDBC (**Java Database Connectivity**) provides four types of drivers to connect Java applications with databases. Each driver type uses a different mechanism to communicate with the database.



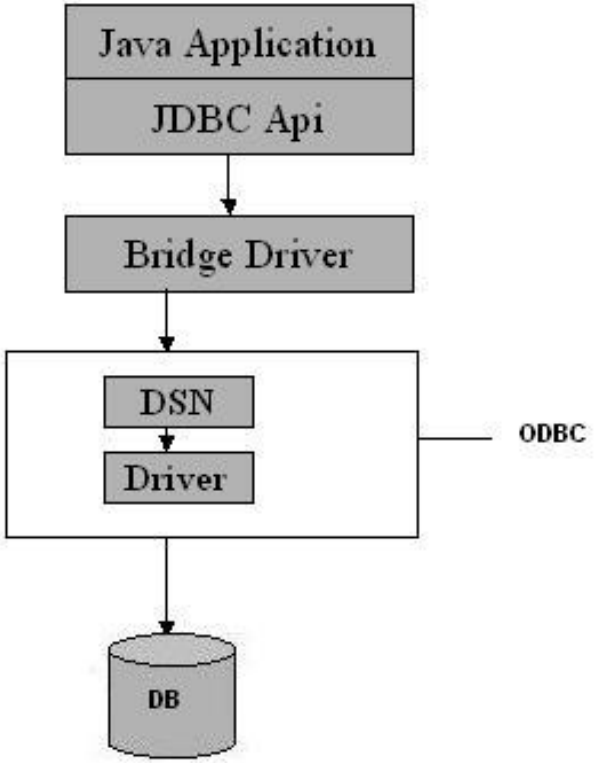
# Types of JDBC Drivers

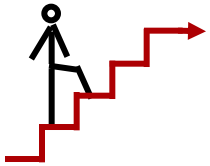
- **Type 1:** JDBC-ODBC Bridge driver (Bridge)
- **Type 2:** Native-API/partly Java driver (Native)
- **Type 3:** All Java/Net-protocol driver (Middleware)
- **Type 4:** All Java/Native-protocol driver (Pure)



# Types of JDBC Drivers

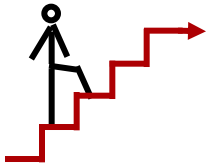
- **Type 1:** JDBC-ODBC Bridge driver (Bridge)
  - Translates JDBC calls into ODBC (Open Database Connectivity) calls.





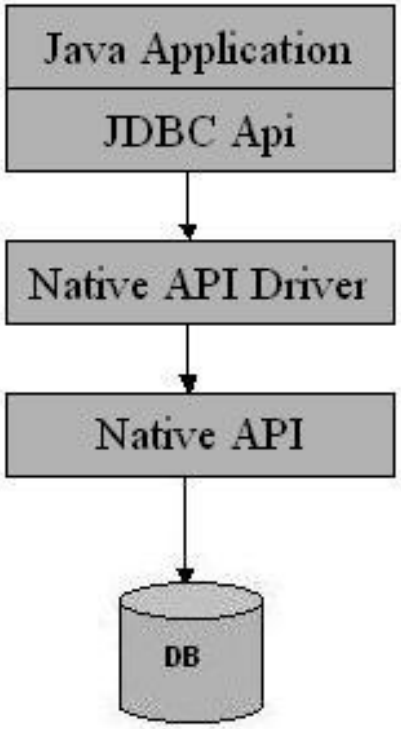
# Types of JDBC Drivers

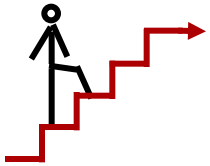
- **Advantages:**
  - Easy to use.
  - Useful for quick prototyping.
- **Disadvantages**
  - Requires ODBC driver on the client machine
  - Platform-dependent.
  - Slower performance.
  - Deprecated in Java 8+.



# Types of JDBC Drivers

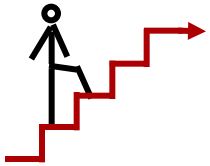
- **Type 2:** Native-API/partly Java driver (Native)
  - Converts JDBC calls into native calls of the database API (e.g., Oracle OCI, DB2 CLI).





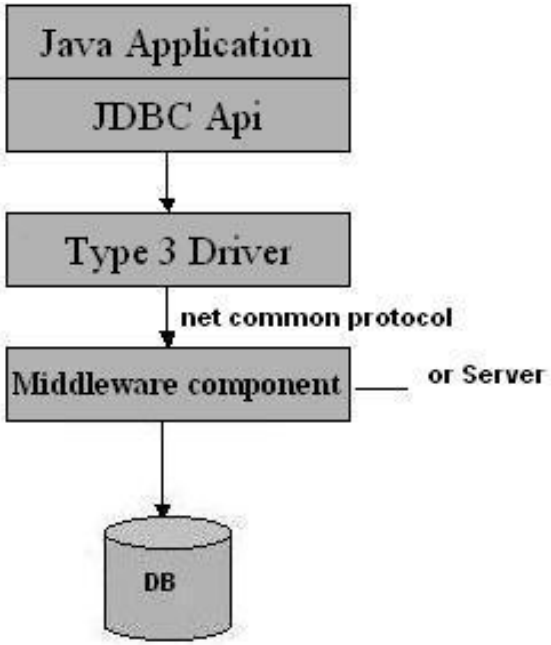
# Types of JDBC Drivers

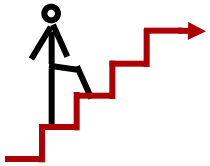
- **Advantages:**
  - Better performance than Type 1.
  - Uses native database libraries.
- **Disadvantages**
  - Requires native library installation.
  - Platform-dependent.
  - Not portable.



# Types of JDBC Drivers

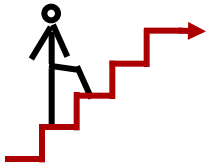
- **Type 3:** All Java/Net-protocol driver (Middleware)
  - Sends JDBC calls to a middleware server which translates them into database-specific calls.





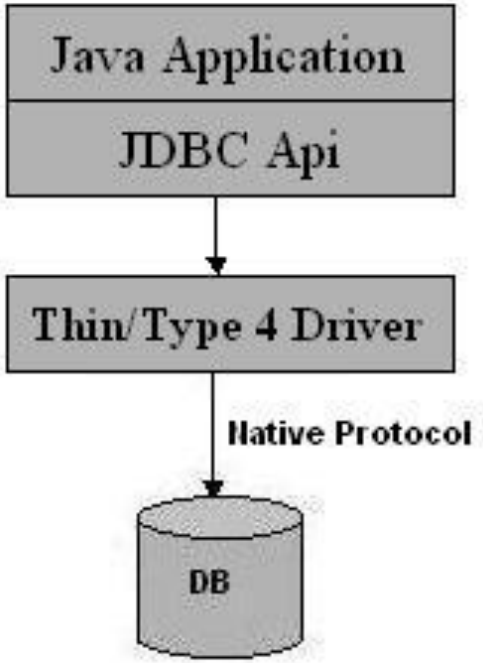
# Types of JDBC Drivers

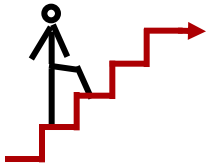
- **Advantages:**
  - No client-side library required.
  - Highly flexible and portable.
  - Suitable for Internet-based applications.
- **Disadvantages**
  - Requires a middleware server.
  - Slightly slower due to additional network layer.



# Types of JDBC Drivers

- **Type 4:** All Java/Native-protocol driver (Pure)
  - Converts JDBC calls directly into the database-specific protocol using pure Java.





# Types of JDBC Drivers

- **Advantages:**
  - 100% Java-based.
  - Platform-independent.
  - Best performance.
  - Most widely used today.
- **Disadvantages**
  - Database-specific (one driver per database).